Predator/Prey Simulation

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Introduction:

Purpose:

This project aims to create an ecosystem where both prey and predators coexist across generations, with genetic mutations that set the scene for natural selection. This is a simple recreation to achieve a simulation that seeks a balanced population while illustrating evolutionary dynamics over time.

Context:

Natural selection is the process that causes species to change over time in response to their environment. In our project, we simulated natural selection using elements of evolutionary algorithms. Specifically, we added a genetic component and implemented a mating operation with mutations, allowing for realistic variations in traits over generations.

Objectives:

- Develop a balanced ecosystem where both predators and prey thrive, forming an ecological pyramid.
- Model realistic differences in genetic values among species that affect their behaviour and survival.
- Demonstrate the impact of mutations and genetic inheritance on natural selection within the simulation.

Simulation Description

Species & Behaviours:

Species	Type	Behaviour	BA	MA	BP	DP	ML	MF	M
Wolf	Predator	Hunts prey aggressively, patrols its territory and reproduces efficiently when encountering a mate.						N/A	
Coyote		Employs opportunistic hunting tactics by scanning adjacent cells for vulnerable prey. It balances moderate aggression with the need to manage hunger and disease.						N/A	
Deer	Prey	Grazes on plants and moves to adjacent free cells to avoid predators. Reproduces upon encountering an opposite-gender							

		mate, while remaining sensitive to hunger and disease.							
Squirrel		Actively forages for food by moving quickly between patches of grass. Reproduces rapidly and uses agile, erratic movements to evade predators.							
Mouse		Scampers swiftly in search of grass to boost its food level, reproduces quickly, and is particularly vulnerable to starvation and disease due to a high metabolic rate.							

* The starting values all animals start with coloured compared to each other:

BA = Breeding Age, the age required to start breeding.
MA = Max Age, the highest age that can be reached.
BP = Breeding Probability, the probability that the species mate and produce offspring.
DP = Disease Probability, the probability that the species contract a disease.
ML = Max Litter size, the maximum amount of offspring at once.
MF = Max Food value, the maximum amount of food points they can store at once.
M = Metabolism, the amount of decrease in the food level of an animal.

** Values indicate:

Red = High value
Amber = Moderate value
Yellow = Low value

Interactions:

Animal Movements & Environment:

At each simulation step, animals execute their act() method, ageing, moving, and updating their hunger levels. They can only move to adjacent free cells. When an animal dies due to starvation, age, or disease, its cell is immediately replaced by grass, ensuring the environment remains dynamic.

Predator-Prey Dynamics:

Predators and prey play distinct roles. Predators search for prey in neighbouring cells, kill live prey to boost their food level and remove them from the simulation. On the other hand, Prey feeds on grass to replenish their food reserves. This food chain dynamic means that the availability of grass directly affects prey survival, which in turn impacts predator populations.

Reproduction and Natural Selection:

Both predators and prey reproduce when conditions allow, though only females give birth. Offspring are placed in free adjacent cells. Each animal's reproductive traits, such as breeding age, lifespan, breeding probability, and litter size, are governed by its gene, a 14-digit string. Mating occurs via a crossover operation with mutation, simulating natural selection and driving population dynamics.

Disease Propagation:

The disease is modelled as a Boolean state. Once an animal becomes infected, it experiences reduced remaining life and can spread the disease to adjacent animals of the same species using the diseaseSpread() method. This mechanism causes localised outbreaks that influence the overall population balance.

Extensions and Implementation Details

Implemented Extensions:

Wolves Pack Hunting and Territorial Behaviour:

The wolves make sure they are near other wolves before gaining a pack advantage to their food levels when hunting in packs, this encourages pack hunting and increased mating. They also tend to mark territories, so they stick to the ground whilst hunting, which also increases mating.

Coyote Opportunistic Scavenging Behaviour:

Coyotes have a chance to increase food levels from newly grown grass showing that recently decaying carcasses could have been present and making them gain advantage due to their scavenging skills. Coyotes can also switch from hunting by themselves or in packs which helps them survive and make crucial survival decisions being very versatile animals.

Deer Flee Response and Grazing Behaviour:

The Deer have a chance to flee from predators if they are able to see them allowing them to survive longer than other prey, as they are more intelligent than others. It has the chance to check if there are any predators nearby and if there aren't it will start to graze but otherwise, if it notices the predator, it will flee away from them.

Squirrel Erratic Movement and Hiding Behaviour:

Squirrels have an unpredictable movement that allows them to move far in a short amount of time in different directions, as well as being able to hide when predators are nearby where it will sometimes decide to just stay put and hide in the grass it is currently on.

Mouse Random Movement and High Mating Behaviour:

Mice tend to move randomly allowing them to be unpredictable and quick. It survives from larger predators by mating much more than other prey.

Grass Pollinating Behaviour:

The Grass also has a chance to pollinate and increase in size by taking free adjacent locations around some grass, this allows for a more realistic approach and helps the prey to not rely on dead animals for grass.

Mouse Random Movement and High Mating Behaviour:

Mice tend to move randomly allowing them to be unpredictable and quick. It survives from larger predators by mating much more than other prey.

Code Structure:

The code is organised into several interrelated modules, each responsible for a specific simulation aspect. This modular structure makes the program both extensible and maintainable. The key components are:

1. Animal Hierarchy:
    - Animal (Abstract Class):
      Serves as the base class for all living entities in the simulation. It encapsulates common

attributes (like life status, location, colour, and a Gene object) and behaviours (such as setting death and disease spread). It also defines abstract methods (e.g., act() and getFoodValue()) that its subclasses must implement.

   - Predator and Prey (Abstract Subclasses):
     These classes extend Animals to model species-specific behaviour.
     - Predator: Implements behaviour for hunting prey, reproducing, ageing, and handling the disease.
     - Prey: Implements behaviours for grazing (feeding on grass), reproduction, and disease handling.
   - Concrete Animal Classes:
     Classes such as Deer, Mouse, Squirrel, Coyote, and Wolf provide concrete implementations of predator and prey behaviours. They customise properties (like lifespan, breeding probability, and food value) while inheriting common functionality from their abstract parents.

2. Plant Representation:
   - Grass:
     Although it extends to Animals, Grass is used to represent plants. It provides a nutritional value via getFoodValue(), which prey animals consume. Grass also includes a reproduction (spreading) mechanism to simulate plant growth and colonisation of free cells in the field.
3. Genetic Mechanism:
   - Gene:
     Encapsulates genetic information that controls an animal's life cycle parameters such as breeding age, lifespan, breeding probability, litter size, disease probability, and metabolism. The gene is represented as a 14-digit string and is used in mating operations with mutation logic to simulate natural selection.
4. Simulation Infrastructure:
   - SimulatorView:
     A JavaFX-based graphical user interface that renders the simulation grid. Each cell is drawn with a colour corresponding to the species occupying it. Incorporates Pause, Resume, and Restart buttons with a Log view and Population Trends graph view to analyse the ecosystem better. It also displays real-time population counts, and generation numbers at the bottom of the screen as well as showing the average genetic values across the population of a specific species in real-time.

This clear separation of concerns, from the underlying animal behaviours and genetic evolution to the environment management and visualisation, ensures that each component can be modified or extended independently, facilitating future enhancements while meeting the coursework requirements.

Conclusion

Summary:

This project reinforced the value of leveraging collaborative tools such as GitHub for coherent teamwork. Using GitHub allowed us to manage version control effectively, coordinate our changes, and merge our work seamlessly within our pair. We learned that following proper coding etiquette such as organising code into clearly defined methods and classes, using descriptive variable names, and employing abstract classes to define shared behaviour leads to cleaner, more maintainable code.